

Primordial Gravitational Waves and CMB Polarization

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The detection of the effects of primordial gravitational waves on the cosmic microwave background (CMB) would provide a window into the early universe, helping reveal physics related to quantum tensor fluctuations and inflation. Doing so requires an understanding of the polarization structure of the CMB and how it can be decomposed into E-modes and B-modes. A key requirement for this decomposition is understanding what primordial processes can produce which modes, as well as how non-primordial contaminants can contribute to the detected spectrum. The unknown strength of primordial gravitational waves and the host of similar-looking contaminants makes detecting these effects a daunting task. Regardless, there are a number of experiments searching for the effects of primordial gravitational waves in the CMB, and even without a detection of the elusive primordial B-modes, these experiments place constraints on the strength of primordial tensor fluctuations.

I. INTRODUCTION

The cosmic microwave background (CMB) has provided a wealth of physics to explore since its discovery in 1964 by Robert Woodrow Wilson and Arno Allan Penzias. With measurements from COBE and FIRAS [1, 2], the CMB was found to be a nearly perfect black body spectrum of about $2.73K$ with small temperature anisotropies. Then DASI and WMAP found the E-modes expected in the polarization of the CMB [3, 4]. The polarization of the CMB can be broken into two different modes, E-modes and B-modes. While the majority of the CMB is composed of E-modes, there is a small component of B-modes which are of particular interest for looking at primordial gravitational waves.

During inflation, quantum fluctuations of the inflation field lead to anisotropies and eventually the formation of large-scale structure. Additionally, quantum fluctuations in the metric tensor itself can produce primordial gravitational waves which then leave their imprint on the polarization of the CMB in the form of B-modes. However, because the power of these B-modes is so small, they are incredibly difficult to detect. Furthermore, different “contaminants” can produce B-modes that look like the B-modes expected from primordial gravitational waves [5]. As such, much effort has been put into understanding these contaminants.

In this work, we explore the polarization of the CMB and its decomposition into E and B-modes, as well as connections to the polarization of gravitational waves in gravitational physics. Then, we discuss how primordial gravitational waves leave their imprint on the polarization of the CMB including their identifying B-mode signature, connecting the gravitational and electromagnetic sectors. Finally, we briefly review some past and current experiments looking for primordial B-modes and the difficulties that come with seeking such an impactful detection.

II. INFLATION

Prior to the theory of inflation, the universe was understood to be described by the Friedmann-Lemaître-Robertson-Walker (FLRW) metric, expressed as

$$ds^2 = -dt^2 + a^2(t)(dr^2 + r^2 d\Omega^2) \quad (1)$$

for a flat universe. The parameter $a(t)$ is the scale factor of the universe, with the expansion rate of the universe (or Hubble parameter) given by $H = \dot{a}/a$. During the radiation and matter dominated eras of the universe, the Hubble distance, H^{-1} , is increasing more rapidly than the scale factor. This means that an observer would see information enter the horizon. However, this raises the question of how did the CMB have nearly the same temperature everywhere if different patches of the sky were not causally connected? There is also the flatness problem, where the universe is observed to have a density parameter of $\Omega = \rho/\rho_c = 1$ which describes a flat universe. This is an unstable fixed point, so any deviations from a flat universe would grow in time. If deviations grow, then how is the universe perfectly balanced to be flat? The absence of monopoles also posed an issue, as they should have been abundantly produced in the early universe [5, 6].

All these issues can be rectified with an inflationary model, where there is an additional scalar field ϕ which begins far away from the minimum of its potential, $V(\phi)$. Such a potential is shown in Fig. 1. The so called inflaton starts away from the minimum of its potential, slowly rolling down the flat portion. During this slow roll, the energy from the inflaton field provides the necessary energy for the rapid expansion of the universe. This rapid expansion can help solve the horizon problem, the flatness problem, and the monopole problem in one shot. The rapid expansion means that patches of the CMB that look causally disconnected were actually causally connected before inflation, allowing the entire CMB to have nearly uniform temperature. As for the flatness problem, the expansion will drive any deviations away from perfect flatness towards zero. The expansion

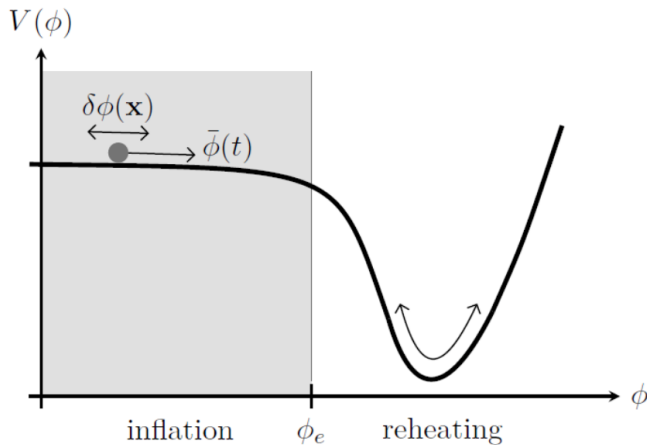


FIG. 1. The slow roll potential of the inflaton field. During the slow roll, quantum fluctuations $\delta\phi(x)$ can nudge the inflaton up and down the potential. Eventually, the universe will enter reheating as the inflaton field exits the slow roll. Figure taken from [7].

will also dilute any monopoles such that their number density is effectively zero. Once the inflaton rolls sufficiently far down its potential the inflationary period will end and the universe enters a period of reheating [5].

During inflation, the inflaton field does not just smoothly roll down its potential. There are small quantum fluctuations in the scalar field, that push the inflaton up and down the potential by small amounts, depicted by $\delta\phi(x)$ in Fig. 1. It is these small fluctuations that lead to the formation of large-scale structure and the temperature anisotropies in the CMB.

A. Primordial Gravitational Waves

During this period of inflation, there were not only the scalar fluctuations of the inflation field, but also quantum tensor fluctuations of the metric tensor itself. The universe could then be described by the metric

$$g_{\mu\nu} = g_{\mu\nu}^{\text{FLRW}} + h_{\mu\nu}, \quad (2)$$

where $h_{\mu\nu}$ encodes the quantum fluctuations on the background FLRW metric. When passed through Einstein's equations, they have wave-solutions, thus producing gravitational waves. An important quantity for characterizing primordial gravitational waves is the tensor-to-scalar ratio,

$$r = \frac{\Delta_t^2}{\Delta_s^2}. \quad (3)$$

This ratio can be thought of as the ratio between the amplitudes of the tensor perturbations and the scalar perturbations. Current bounds put this ratio at $r \lesssim 0.04$ [8]. There are good measurements of Δ_s^2 from CMB temperature anisotropies, but r is left as a free parameter. This

is because r can depend on the exact theory of inflation. Different shapes for the inflaton potential can lead to different values of r . Thus, constraining this parameter is important for constraining inflationary theories.

There is also an interesting heuristic connection between primordial gravitational waves and Hawking radiation. Say there is a black hole and it has a horizon past which information disappears. When quantum mechanics at the horizon is considered, however, one finds that it will emit radiation. Normally this is thought of as electromagnetic radiation, but this radiation can be any radiation field with massless quanta like gravitational waves. This is like the reverse case of what happens in the an FLRW universe with an accelerated expansion. There is a horizon past which information disappears, however, this time the observer is inside the horizon rather than outside, as depicted in Fig. 2. Just like the black hole horizon, this horizon also emit gravitational waves. During inflation, these gravitational waves are produced at the horizon, leaving a primordial gravitational wave background after inflation is over [5].

To better understand primordial gravitational waves, one would hope for some sort of detection of these gravitational waves or their effects. Fortunately, primordial gravitational waves will leave their imprint on the CMB polarization pattern. Thus, we now turn to the polarization of the CMB.

III. CMB POLARIZATION

A. Stokes Parameters

To describe the polarization of the CMB, first let us consider a monochromatic electromagnetic wave of frequency ω_0 propagating in the $\hat{z}$ direction. The components of the electric field of such a wave can be written as

$$E_x = a_x \cos(\omega_0 t - \phi_x(t)) \quad (4)$$

$$E_y = a_y \cos(\omega_0 t - \phi_y(t)). \quad (5)$$

The Stokes parameters, given by I, Q, U , and V are then defined as

$$I \equiv \langle a_x^2 \rangle + \langle a_y^2 \rangle \quad (6)$$

$$Q \equiv \langle a_x^2 \rangle - \langle a_y^2 \rangle \quad (7)$$

$$U \equiv \langle 2a_x a_y \cos(\phi_x - \phi_y) \rangle \quad (8)$$

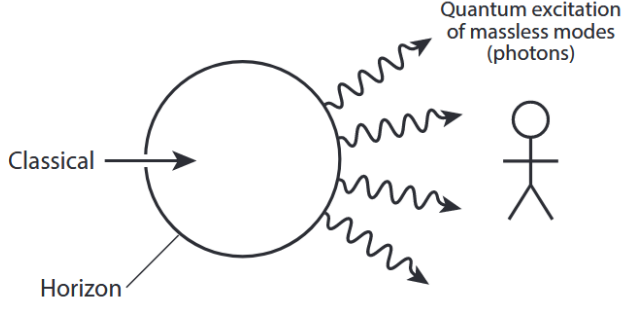
$$V \equiv \langle 2a_x a_y \sin(\phi_x - \phi_y) \rangle \quad (9)$$

where $\langle \rangle$ is a time average [9]. The parameter I corresponds to the intensity of the light. Looking at Fig. 3, we see that the Q parameter measures polarization relative to $\hat{x}$ and $\hat{y}$, while the U parameter measures polarization at a 45° offset. The parameter V on the other hand, encodes the circular polarization.

An important thing to note about the Q and U Stokes parameters is that they are coordinate dependent. Depending on how the coordinate axes are chosen, they will

Black hole analogy

a Black hole



b Inflationary Universe

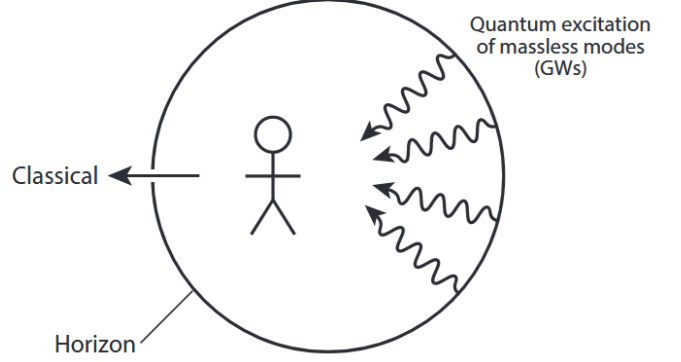


FIG. 2. Analogy between Hawking radiation from a black hole's horizon and primordial gravitational waves from the horizon of an inflationary universe. Figure taken from [5].

100% Q	100% U	100% V
+Q $Q > 0; U = 0; V = 0$ (a)	+U $Q = 0; U > 0; V = 0$ (c)	+V $Q = 0; U = 0; V > 0$ (e)
-Q $Q < 0; U = 0; V = 0$ (b)	-U $Q = 0; U < 0; V = 0$ (d)	-V $Q = 0; U = 0; V < 0$ (f)

FIG. 3. Stokes parameters encoding polarization. Parameters Q and U encode linear polarization while V encodes circular polarization. Figure taken from [10].

change. Specifically, if the coordinate system is rotated by an angle α , these Stokes parameters will transform as

$$Q' = Q \cos(2\alpha) + U \sin(2\alpha) \quad (10)$$

$$U' = -Q \sin(2\alpha) + U \cos(2\alpha). \quad (11)$$

This form of transformation is exactly how a symmetric, traceless, rank-2 tensor transforms [11].

B. Polarization Tensor

Because the Stokes parameters transform in this way, the polarization information from the Stokes parameters can be compacted into the polarization tensor, $P_{\mu\nu}$. As an observer, the CMB is seen as a 2-sphere, S^2 , typically in spherical-polar coordinates (θ, ϕ) . In these coordinates, the polarization tensor in each direction $\hat{n}$ is

given by

$$P_{\mu\nu}(\hat{n}) = \frac{1}{2} \begin{pmatrix} Q(\hat{n}) & -U(\hat{n}) \sin \theta \\ -U(\hat{n}) \sin \theta & -Q(\hat{n}) \sin^2 \theta \end{pmatrix}. \quad (12)$$

The processes that create the CMB, further discussed in Sec. IV, cannot produce any circular polarization, hence the V Stokes parameter is not considered [11].

Even though all the polarization information is now compacted into a singular object, it still has the drawback of being coordinate dependent. Thus if there are different detectors using different coordinate systems, their polarizations tensor will be different as well. To construct coordinate invariant quantities, the polarization tensor is decomposed in terms of basis functions, namely the tensor spherical harmonics.

C. Tensor Spherical Harmonics

Whenever there is a physical observable like a polarization tensor, it can often be useful to break it up into more physically relevant components. Take for example a scalar field $\varphi(\theta, \phi)$ on S^2 . The natural way to decompose $\varphi(\theta, \phi)$ is in terms of spherical harmonics,

$$\varphi(\theta, \phi) = \sum_{\ell, m} A_{\ell m} Y_{\ell m}(\theta, \phi). \quad (13)$$

Such a decomposition allows for easy projections through orthogonality conditions, easy identification of the multipolar structure, and much more. Such a decomposition of a scalar field can naturally be extended to tensor fields with the use of tensor spherical harmonics. Here, we specialize to the case where the tensor to be decomposed is a symmetric, traceless, rank-2 tensor on S^2 . Such are the conditions on the polarization tensor, and they produce many simplifications to the tensor harmonics used here. However, the tensor harmonics can be generalized. Using

the notation of Ref. [12], there are two “classes” of tensor spherical harmonics, those that have even (polar) parity, and those that have odd (axial) parity. The even parity tensor spherical harmonics $Z_{\mu\nu}^{(\ell m)}$ are given by

$$Z_{\mu\nu}^{(\ell m)}(\theta, \phi) = \nabla_\mu \nabla_\nu Y^{(\ell m)}(\theta, \phi) + \frac{\ell(\ell+1)}{2} \sigma_{\mu\nu} Y^{(\ell m)}(\theta, \phi) \quad (14)$$

where $\sigma_{\mu\nu}$ is the metric of S^2 ,

$$\sigma_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}. \quad (15)$$

The odd parity tensor spherical harmonics $X_{\mu\nu}^{(\ell m)}$ are given by

$$X_{\mu\nu}^{(\ell m)}(\theta, \phi) = \nabla_{(\mu} \epsilon_{\nu)}^{\rho} \nabla_\rho Y^{(\ell m)}(\theta, \phi) \quad (16)$$

where $\epsilon^{\mu\nu}$ is the 2D Levi-Civita tensor on S^2 , given by

$$\epsilon_{\mu\nu} = \begin{pmatrix} 0 & \sin \theta \\ -\sin \theta & 0 \end{pmatrix}. \quad (17)$$

Under the parity operator $\hat{\mathcal{P}}$, these tensor spherical harmonics behave as

$$\hat{\mathcal{P}} [Z_{\mu\nu}^{(\ell m)}(\theta, \phi)] = (-1)^\ell Z_{\mu\nu}^{(\ell m)}(\theta, \phi), \quad (18)$$

$$\hat{\mathcal{P}} [X_{\mu\nu}^{(\ell m)}(\theta, \phi)] = -(-1)^\ell X_{\mu\nu}^{(\ell m)}(\theta, \phi). \quad (19)$$

This parity operator takes $\theta \rightarrow \pi - \theta$ and $\phi \rightarrow \phi + \pi$ which amounts to a reflection over S^2 , i.e. $\hat{r} \rightarrow -\hat{r}$. From this behavior, $Z_{\mu\nu}^{(\ell m)}(\theta, \phi)$ having even parity and $X_{\mu\nu}^{(\ell m)}(\theta, \phi)$ having odd parity becomes apparent. While $Z_{\mu\nu}^{(\ell m)}(\theta, \phi)$ is invariant under the parity transformation up to a factor of $(-1)^\ell$, $X_{\mu\nu}^{(\ell m)}(\theta, \phi)$ has an additional minus sign. With these definitions in hand, the polarization tensor can be decomposed as

$$P_{\mu\nu} = \sum_{\ell, m} \left(a_{\ell m}^E Z_{\mu\nu}^{(\ell m)} + a_{\ell m}^B X_{\mu\nu}^{(\ell m)} \right), \quad (20)$$

where there is an even parity component and odd parity component. As discussed in the next section, these correspond to the so-called E and B-modes, hence the notation on the coefficients for each parity harmonic.

D. E and B-Modes

It is this decomposition that then defines what E-modes and B-modes are. E-modes correspond to the even-parity tensor harmonics, $Z_{\mu\nu}^{(\ell m)}$, while B-modes correspond to the odd-parity tensor harmonics, $X_{\mu\nu}^{(\ell m)}$. The advantage of such a decomposition is that no matter the coordinates chosen by a detector, E-modes will always be E-modes, and B-modes will always be B-modes. A change of coordinates cannot mix the two parities.

This fact allows for certain quantities to be truly invariant of the chosen coordinates. One of the vital invariants is the power spectrum for the different polarizations. From Ref. [11], the power spectrum of each polarization, given by C_ℓ^{EE} and C_ℓ^{BB} , are

$$\langle a_{\ell m}^{E*} a_{\ell' m'}^E \rangle = C_\ell^{EE} \delta_{\ell\ell'} \delta_{mm'}, \quad (21)$$

$$\langle a_{\ell m}^{B*} a_{\ell' m'}^B \rangle = C_\ell^{BB} \delta_{\ell\ell'} \delta_{mm'}. \quad (22)$$

Although we now have the mathematical description of E and B-modes, it is good to have a physical intuition for what the different modes actually looks like. Looking at Fig. 4, the top is a polarization spectrum of completely E-modes while the bottom is completely B-modes. E-modes, which have even parity, take the form of either radial or tangential lines. The B-modes however, take the form of either left-handed or right-handed spirals. Here, we see the physical manifestation of the even parity. If either the radial or tangential pattern is reflected over some line, it remains unchanged. When the B-mode patterns are reflected over some line, a right-handed spiral turns into a left-handed spiral and vice-versa. This is the physical manifestation of B-modes having odd parity.

E. Relation to Gravitational Wave Polarization

Although it is customary to decompose the CMB polarization tensor in terms of these even and odd parity tensor harmonics, this concept is closely related to the spin-weighted spherical harmonics more common in gravitational physics, particularly used for gravitational wave polarization from compact binary coalescences. The standard way to look at these gravitational waves is in the transverse-traceless gauge where the metric perturbation from the gravitational waves $h_{\mu\nu}$ is written as

$$h_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (23)$$

Because of the many zeros within the metric, the non-trivial components form a symmetric, traceless, rank-2 tensor, just like the CMB polarization tensor. However, instead of decomposing this in terms of tensor spherical harmonics, it is typically decomposed in terms of spin-weighted spherical harmonics. The spin-weighted harmonics are essentially a projection of the tensor harmonics into a particular basis. This basis is expressed as a dyad of two null basis vectors in the tangent plane, m_μ and $\bar{m}_\mu$ which are complex conjugates of one another. They are given by

$$m_\mu = \frac{1}{\sqrt{2}}(1, i \sin \theta) \quad (24)$$

$$\bar{m}_\mu = \frac{1}{\sqrt{2}}(1, -i \sin \theta). \quad (25)$$

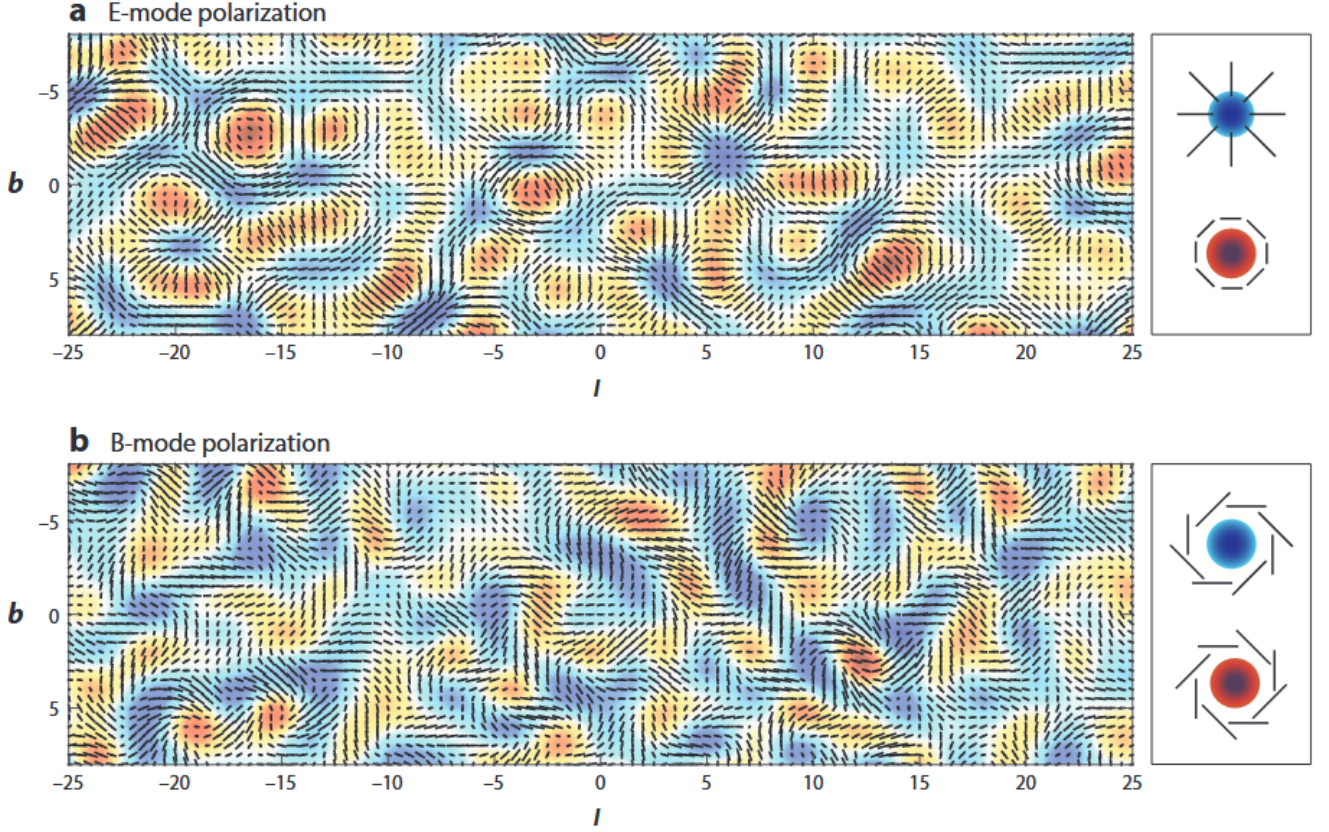


FIG. 4. Polarization patterns composed of only E-modes (top) and of only B-modes (bottom). E-modes can take either a radial or tangential pattern while B-modes have either a right-handed or left-handed spiral pattern. Figure taken from [5].

Together, they also form the metric of the 2-sphere,

$$\sigma_{\mu\nu} = m_{(\mu} \bar{m}_{\nu)}. \quad (26)$$

Then the spin-weighted harmonics of spin-weight $s = \pm 2$ can be computed from the tensor harmonics (up to a normalization) via a projection into the dyad basis,

$$+_2 Y^{\ell m}(\theta, \phi) \propto m^\mu m^\nu (Z_{\mu\nu}^{\ell m} - i X_{\mu\nu}^{\ell m}) \quad (27)$$

$$-_2 Y^{\ell m}(\theta, \phi) \propto \bar{m}^\mu \bar{m}^\nu (Z_{\mu\nu}^{\ell m} + i X_{\mu\nu}^{\ell m}). \quad (28)$$

Spin-weighted harmonics with $s = \pm 1$ can be formed from a similar projection of the vector harmonics. The $s = 0$ spin-weighted harmonics are identically the normal spherical harmonics.

With this relationship in hand, it is helpful to understand the physical meaning of spin-weight. The spin-weight describes how a quantity η behaves under a rotation of the null basis vectors by an angle ψ , $m_\mu \rightarrow e^{i\psi} m_\mu$ and $\bar{m}_\mu \rightarrow e^{-i\psi} \bar{m}_\mu$. Under such a transformation, the quantity η of spin-weight s transforms as

$$\eta \rightarrow e^{is\psi} \eta. \quad (29)$$

In this context, it becomes clear that the tensor harmonics $Z_{\mu\nu}^{\ell m}$ and $X_{\mu\nu}^{\ell m}$ have some pieces that transform as $s = +2$ objects, and other pieces that transform as

$s = -2$. The respective complex linear combinations to form the spin-weighted harmonics then isolate either the $s = +2$ or $s = -2$ portions. In a similar sense, the spin-weighted harmonics has some pieces that have even parity and other pieces that have odd parity. Thus it can be helpful to think of the tensor harmonics as parity eigenstates while the spin-weighted harmonics are spin-weight eigenstates. Then with particular linear combinations and projections into the dyad basis, one can go between the two.

With the mathematical description of spin-weighted harmonics in hand, we can now look at how they are used to decompose gravitational wave polarization. Gravitational waves are normally decomposed as

$$h_+ - ih_\times = \sum_{\ell, m} h_{\ell m} {}_{-2}Y_{\ell m} \quad (30)$$

where the $h_{\ell m}$ is the coefficient for each harmonic. This decomposition describes left circularly-polarized gravitational waves, and the standard plus and cross polarization can then be read off from the real and imaginary parts.

It is quite interesting to note the connection between the tensor harmonics and spin-weighted harmonics, as well as the differences between how cosmologists and gravitational physicists look at polarization. The dif-

ferences are physically motivated by the specific physics within each subfield. For cosmologists, the even and odd parity tensor harmonics have physical relevance in terms of what produced the polarization. E-modes come from Compton scattering, whereas B-modes come from primordial gravitational waves (or other contaminants). However, in gravitational physics, plus and cross polarization is more relevant since they are more concerned with how a detector sees gravitational waves, and the physical origin of even and odd parity polarization is less relevant.

IV. COMPTON SCATTERING

With the polarization tensor decomposed into E and B-modes, an understanding of what processes produce which modes is important. Once the universe has sufficiently cooled, interactions between photons and electrons through Compton scattering become more infrequent. Consequently, we get a surface of last scattering, after which scattered photons free stream from this surface to us today, forming the CMB. However, the process of Compton scattering leaves an imprint in the CMB in the form of polarization. First, recall that there is a quadrupolar temperature anisotropy at the surface of last scattering. Now imagine an electron where there is incoming light from above the electron and from the left, as depicted in Fig. 5. Say that the light coming from the top is hotter than the light coming from the left, a manifestation of the temperature anisotropy. As this light hits the electron and scatters, the resulting photon will have a linear polarization that reflects this temperature anisotropy. In particular, there will be a net linear polarization that is aligned with the cold axis [13].

Notably, this process is invariant under the parity transformation. If we take the electron to be the center of our coordinate system, and then reflect everything over the 2-sphere, we see that the same polarization of light would result. This is very important because this means that the process of Compton scattering has even parity and thus produces only E-modes. Therefore, if the light from in the CMB is completely from Compton scattering and travels unaffected to an observer today, it would be composed completely of E-modes. In more mathematical terms,

$$P_{\mu\nu}^{\text{Compton}} = \sum_{\ell, m} a_{\ell m}^E Z_{\mu\nu}^{(\ell m)}. \quad (31)$$

V. POLARIZATION OF PRIMORDIAL GRAVITATIONAL WAVES

The primordial gravitational waves described in Sec. II A have an effect on the CMB as well. However, primordial gravitational waves are perturbations of the

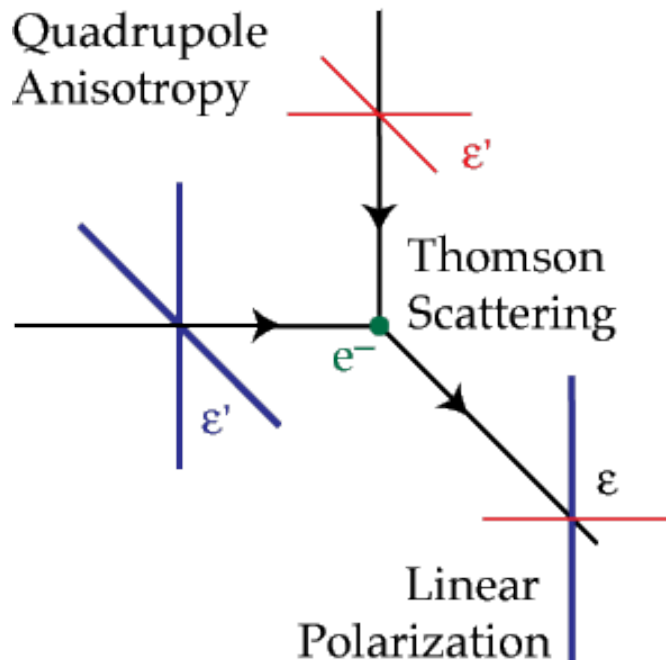


FIG. 5. Linear polarization from Compton (relativistic Thomson) scattering. The temperature quadrupole anisotropy means the light hitting the electron from above is hotter than the light hitting the electron from the left. As a result, the scattered light coming out of the page is linearly polarized. Figure taken from [13].

metric tensor, so that raises the question of how does this create a polarization pattern within the CMB. The mathematical derivation of how metric perturbations result in changes in polarization is not straightforward, but is done in [5]. The main result is that the metric perturbations cause changes to the Stokes parameters Q and U , relationships dependent on the angle between the propagation direction of the gravitational wave and the photon trajectory, as well as the azimuthal angle of the photon's trajectory. To see this effect in practice, take a singular gravitational wave traveling in the $\hat{z}$ direction. The corresponding polarization pattern is shown in Fig. 6. From this figure, we see that this gravitational wave produces these swirling polarization patterns, the signature of a B-mode polarization. Note that this pattern contains both E and B-modes, as the polarization pattern contains both types of modes.

The mixing of E and B-modes from a gravitational wave can be seen if we look at its polarization tensor as well. Considering a single gravitational wave of $+$ polarization, wavelength k , and propagating in the $\hat{z}$ direction. The corresponding polarization tensor is given by

$$P_{k,+}^{ab}(\theta, \phi) = \frac{T_0}{4\sqrt{2}} \sum_{\ell} (2\ell + 1) P_{\ell}(\cos \theta) \tilde{\Delta}_{Q\ell}$$

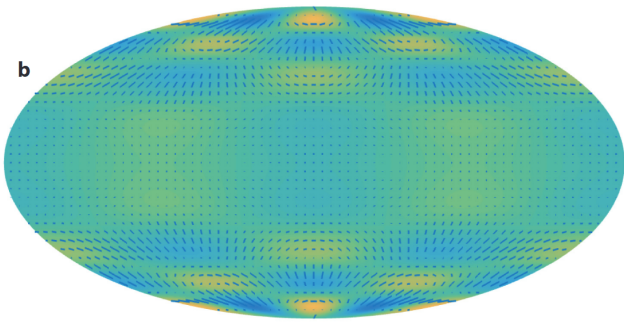


FIG. 6. Temperature-polarization pattern from a singular gravitational wave. Color gradient shows the quadrupolar variation in the temperature, and lines show the polarization. The polarization contains both E and B-modes. Figure taken from [5].

$$\times \begin{bmatrix} (1 + \cos^2 \theta) \cos 2\phi & 2 \cot \theta \sin 2\phi \\ 2 \cot \theta \sin 2\phi & -(1 + \cos^2 \theta) \csc^2 \theta \cos 2\phi \end{bmatrix}, \quad (32)$$

where T_0 is the mean temperature, $\tilde{\Delta}_{Q\ell}$ is the transfer function for Q polarization, and ϕ is the azimuthal angle [5]. Intuitively, the transfer function tells us how efficiently a gravitational wave mode of wavelength k produces Q -polarization at angular scale ℓ . This polarization tensor has both E and B-modes, which can be shown by enacting the parity transform $\hat{P}$ on this tensor and finding it does not have definite parity. If it does not have definite parity, it must be some combination of even-parity and odd-parity component, i.e. a combination of both E and B-modes. Because primordial gravitational waves make primordial B-modes and are thought to be the only source of such primordial modes, a detection of primordial B-modes within the CMB would be direct evidence for primordial gravitational waves.

VI. DETECTION PROSPECTS

A. Contamination to B-modes

Although it is thought that primordial gravitational waves are the only primordial contributors to B-modes within the CMB, there can be a lot of “contaminants” that can create B-modes as the light from the surface of last scattering makes its way to detectors today. The largest of these effects is lensing-induced B-modes. As CMB photons travel from the surface of last scattering, they will pass by large-scale structure such as galaxies and galaxy clusters. When this happens, the photons will get weakly gravitationally lensed which introduces a cosmic shear [5]. Without doing a deep dive into the math, this lensing effectively turns E-modes from Compton scattering into B-modes that will look very similar to those from primordial gravitational waves.

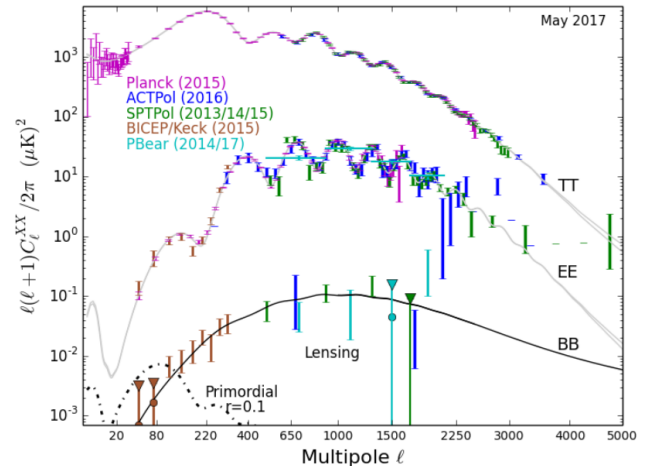


FIG. 7. Power spectrum of the CMB as a function of angular scale ℓ . Temperature (TT), E-modes (EE), and B-modes (BB) are shown. B-modes are broken into those seen from lensing and those predicted from primordial gravitational waves with $r = 0.1$. Note that more current constraints have $r \lesssim 0.04$, which would lower this line. Figure taken from [14].

To ensure that lensing-induced B-modes are not mistaken for primordial B-modes, there needs to be a good understanding of what they will look like. Using models for the angular structure of the lensed B-modes, there have been detections of these B-modes, shown in Fig. 7. This figure plots a scaling of the power spectrum described in Sec. III D as a function of the angular scale ℓ . Lensed B-mode detections are quite amazing considering how much weaker the B-modes are than the temperature anisotropies and the E-modes. Note that the lensed B-modes have a distinctly different angular structure from the primordial B-modes, something that can help disentangle the two.

In addition to the gravity of galaxies turning E-modes into B-modes, the galaxies can simply produce B-modes themselves. Galactic synchrotron radiation lies within the frequency range of the CMB, so understanding this foreground contamination is essential for being able to filter it out of CMB signals. Dust surrounding galaxies can also be problematic. As dust aligns with galactic magnetic fields, it can emit strongly polarized light, again in the frequency range of the CMB. Because these signals from dust have complex dependence on composition, shape, and size of the grains, it is hard to model accurately without significant guesswork [5]. As such, more work is needed to understand these foregrounds.

B. Current Experiments

There are a number of current and past experiments that have looked for and are looking for B-modes from primordial gravitational waves. The most infamous of

which is BICEP2, which announced they had detected primordial B-modes in 2014. However, subsequent analyses including data from the *Planck* satellite showed that Galactic dust could be responsible for the B-mode signal. They had not properly accounted for the B-modes from foreground, showing the difficulty and importance of accurately modeling, accounting for, and filtering out possible B-mode contaminants.

Meanwhile, experiments like POLARBEAR and the Atacama Cosmology Telescope have been able to find lensing-induced B-modes, matching theoretical predictions for how large-scale structure can turn E-modes into B-modes. While not the smoking gun of primordial B-modes, it is still vital science to understand different sources of B-modes and large-scale structure. Today, experiments like Simons Observatory, Cosmology Large Angular Scale Surveyor, and POLARBEAR are all actively searching for primordial B-modes. These experiments have increased sensitivity, hopefully enough to detect primordial B-modes soon.

VII. CONCLUSION

In this exploration of primordial gravitational waves and their relation to CMB polarization, we have seen how

quantum tensor fluctuations during inflation produce primordial gravitational waves. The relative strength of these gravitational waves, described by the tensor-to-scalar ratio r , can vary depending on the inflationary theory, but is constrained to $r \lesssim 0.04$ today [8]. By developing an understanding of how to decompose the CMB polarization into E and B-modes, we see how the Compton scattering that forms the CMB can only produce E-modes. However, primordial gravitational waves generate E and B-modes, leaving their imprint on the polarization of the CMB. The decomposition of the polarization tensor into tensor spherical harmonics also has close conceptual ties with the way the polarization of gravitational waves is decomposed. We explored the connection between the tensor spherical harmonics conventionally used for CMB polarization and the spin-weighted spherical harmonics conventionally used for gravitational wave polarization.

Finally, we discuss the detection prospects for primordial gravitational waves, including the false detection of primordial B-modes by BICEP2 in 2014. There are a number of current experiments searching for primordial B-modes, with ever-increasing sensitivities. With the improvement of these experiments, hopefully they can finally detect primordial B-modes, possibly shedding light in new inflationary physics.

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